# AURA

## Automated User Resource Analyzer

AI-Powered Predictive Maintenance for the ISS Environmental Control & Life Support System (ECLSS)

*4 stacked ML models · Real-time anomaly detection · Fault classification · AI remediation*

Python 3.11 · FastAPI · PyTorch · scikit-learn · SQLite · WebSocket · Isolation Forest · Random Forest · LSTM · DQN

| <1s | 11 | 8 | 20+ | 7 | 4 |
|---|---|---|---|---|---|
| Tick Interval | Remediation Actions | Fault Types | Sensor Parameters | ISS Locations | ML Models |

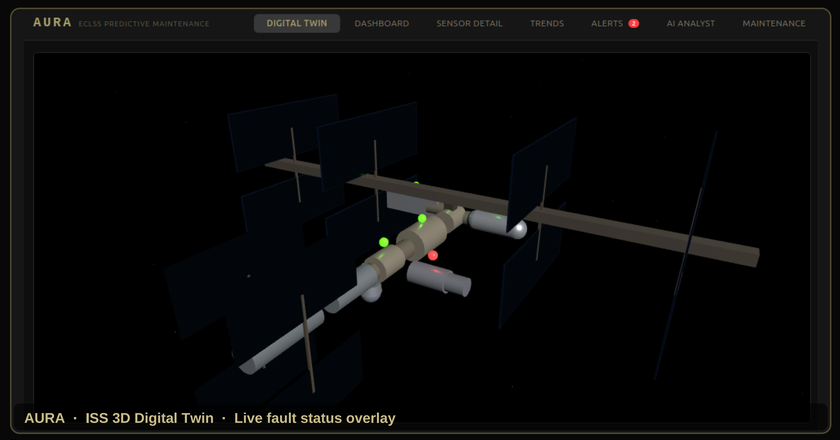
## WHAT IS AURA?

AURA is a full-stack real-time dashboard that monitors the ISS Environmental Control and Life Support System across 7 station modules. It generates physics-based sensor data, runs 4 stacked ML models on every tick, and streams results live via WebSocket — giving crew and ground control early warning of faults before they become mission-critical failures.

## ML INFERENCE PIPELINE

1. **SENSOR SIM** — Physics-based ISS correlated data
2. **ISOLATION FOREST** — Anomaly detection unsupervised
3. **RANDOM FOREST** — 8-class fault identification
4. **LSTM NETWORK** — Failure prob & RUL prediction
5. **DEEP Q-NET** — Optimal remediation action selection

## DASHBOARD




AURA · ISS 3D Digital Twin · Live fault status overlay

## FAULT TYPES DETECTED

- **Cabin Leak** — Cabin pressure / N2 drop — structural seal breach
- **O2 Generator Failure** — OGS electrolysis cell stack degradation
- **O2 Leak** — O2 plumbing fitting / valve seal failure
- **CO2 Scrubber Failure** — CDRA zeolite sorbent bed exhausted
- **CHX Failure** — CCAA heat exchanger hydrophilic coating loss
- **Water Processor Failure** — WPA sieve / MF bed contamination
- **Trace Contaminant Saturation** — TCCS charcoal bed — NH3 / CO breakthrough
- **NH3 Coolant Leak** — EATCS toxic ammonia loop breach

## MACHINE LEARNING STACK

### Isolation Forest

*Anomaly Detection · Layer 1*

- Unsupervised — trained on nominal ISS data only; no labeled faults required
- Isolates anomalies by random partitioning; shorter path = stronger outlier
- 5-consecutive-tick filter: P(5 FPs at 3% FPR) ≈ 2 × 10■■ per day
- Per-location 300 s cooldown prevents repeat alert fatigue
- IF label passed as binary feature into every downstream model layer

### Random Forest

*Fault Classification · Layer 2*

- Multi-class classifier: predicts which of 8 fault types is currently active
- Outputs a full probability distribution across all fault classes each tick
- ≥ 60 % confidence activates the live fault indicator in the UI
- ≥ 90 % confidence latches fault display until crew manually resolves it
- Trained on physics-derived sensor drift signatures per fault type

### LSTM Neural Network

*Failure Prediction · Layer 3*

- Long Short-Term Memory captures temporal degradation in sensor sequences
- Outputs failure probability [0 – 1] and Remaining Useful Life in hours
- Per-location rolling buffer resets on fault injection for a clean signal
- Enables proactive maintenance hours before fault reaches critical state
- failure_prob and RUL forwarded as state inputs to the DQN layer below

### Deep Q-Network (DQN)

*Remediation Recommendation · Layer 4*

- Reinforcement learning: maps full system state → optimal repair action
- 11 discrete actions: sealant, O2 valve isolation, EVA prep, bed R&R, …
- State vector: sensor readings, anomaly score, RF class, LSTM probability
- Outputs action index + confidence from Q-value softmax distribution
- Recommendations displayed live per ISS module in the dashboard UI

**O2 Generator Failure** — CONDITION · NOMINAL
*Oxygen Generation Assembly (OGA/OGS)*
4.1% life used — 17.7 wk remaining

**Water Processor Failure** — CALENDAR · NOMINAL
*Water Recovery Assembly (WPA/WRS)*
3.3% life used — 22.2 wk remaining

## KEY FEATURES

- **Real-Time WebSocket Streaming** — Sub-second sensor ticks broadcast to all connected browsers via FastAPI WebSocket.
- **Fault Latch Mechanism** — At 90 % RF confidence the fault is pinned per-location until crew explicitly resolves it.
- **LSTM + DQN Integration** — Failure probability and RUL from LSTM feed directly into the DQN state vector.
- **Smart Alert Debouncing** — 5-tick threshold + 5 min cooldown: P(false sustained alarm at 3 % FPR) ≈ 2×10■■/day.
- **AI Narrative Analyst** — Ollama / Mistral LLM generates natural-language diagnostics per location on demand.
- **Trend Detection** — Sen's slope + CUSUM detects gradual parameter drift before the anomaly threshold.
- **Maintenance Scheduler** — MTBF-based replacement and drift-based calibration schedules for all 8 subsystems.
- **Digital Twin Floorplan** — Live SVG floorplan of the ISS overlaid with per-location fault states and severity.

## ALERT SYSTEM DESIGN

| | |
|---|---|
| IF flags anomalous reading | 1 TICK |
| N = 5 consecutive anomalous ticks | TRIGGER |
| RF classifies fault at ≥ 60 % | LIVE UI |
| RF confidence reaches ≥ 90 % | LATCH |
| Alert fired — 5 min cooldown reset | CRITICAL |

## DATABASE SCHEMA (SQLite)

**sensor_readings**
- id INTEGER PK
- location TEXT
- timestamp TEXT
- data JSON

**sensor_labels**
- reading_id FK
- if_label INT
- rf_classification JSON

**alerts**
- id PK
- location / severity
- fault_type
- acknowledged

**active_faults**
- location PK
- fault_name TEXT
- start_time TEXT

CRITICAL — O2 Leak (93%) — Joint Airlock
2026-04-17T12:52:19.619313

**TECHNOLOGY STACK**

**Backend:** Python 3.11 · FastAPI 0.115 · Uvicorn · asyncio · WebSocket · Pydantic
**ML / AI:** PyTorch ≥2.0 · scikit-learn ≥1.5 · Isolation Forest · Random Forest · LSTM · DQN
**Database:** SQLite3 · JSON column storage · 5,000-row rolling buffer · WAL journaling
**Frontend:** Vanilla JS · CSS custom properties · WebSocket client · SVG ISS digital twin
**AI Analyst:** Ollama (Mistral) · on-demand natural-language diagnostics per location via REST
**Deploy:** Cross-platform: Windows .bat / Linux .sh · Zero Docker dependency

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*Computer Science · Database Systems · Final Project 2026*

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